



BIOMECHANICAL REVIEW OF RUNNING TECHNIQUE

NAME: Stefano Villa
AGE: 29

HEIGHT 181 cm
WEIGHT 66 kg

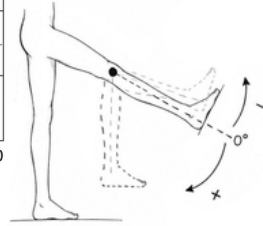
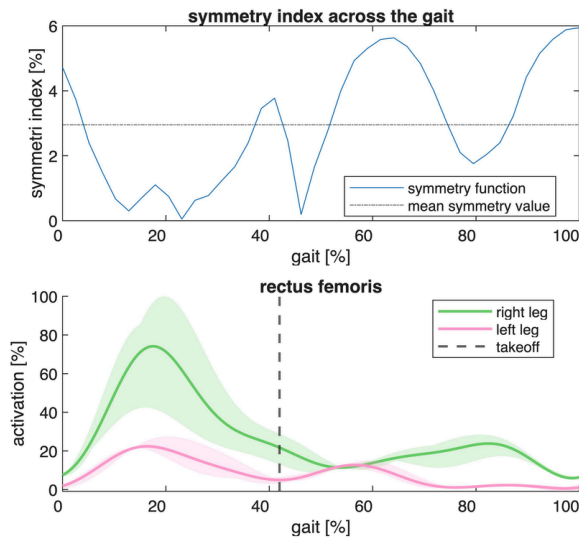
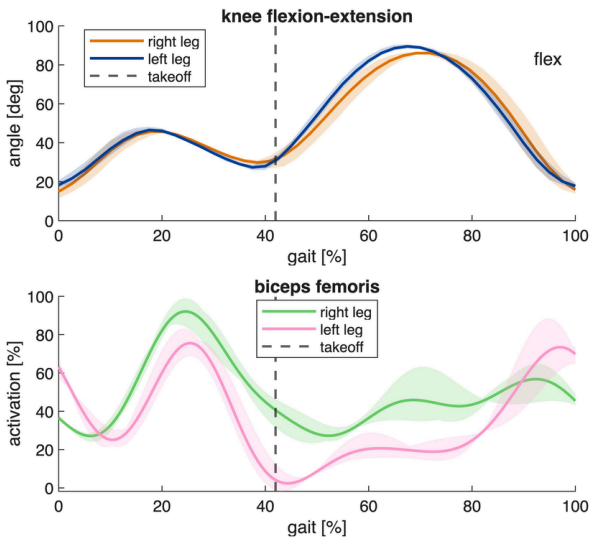
VO2MAX: 68.7 ml/Kg*min
KM/WEEK: 80

INJURIES: Second-degree ankle sprain, meniscal injury

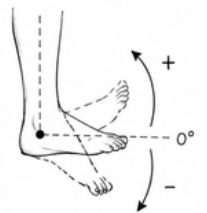
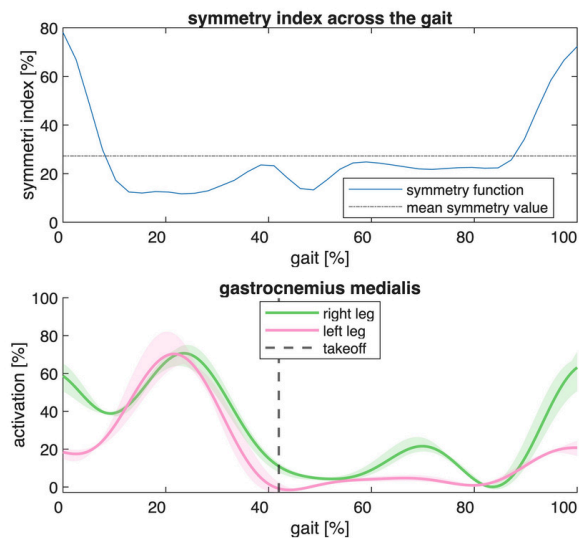
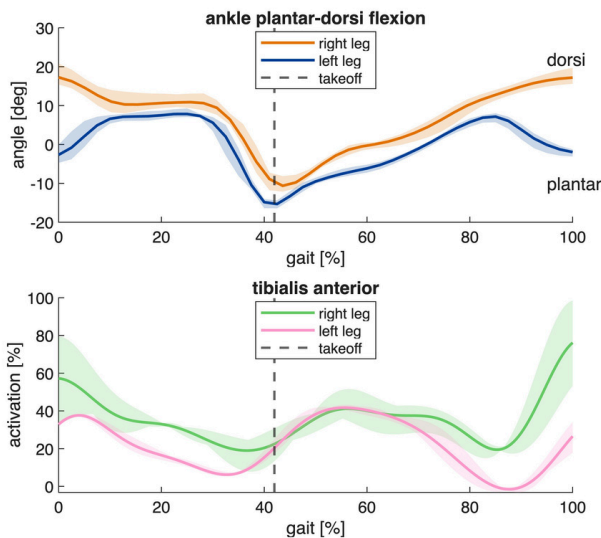
ASYMMETRIES

Evaluation of left-right asymmetry indices, focusing on kinematic parameters.

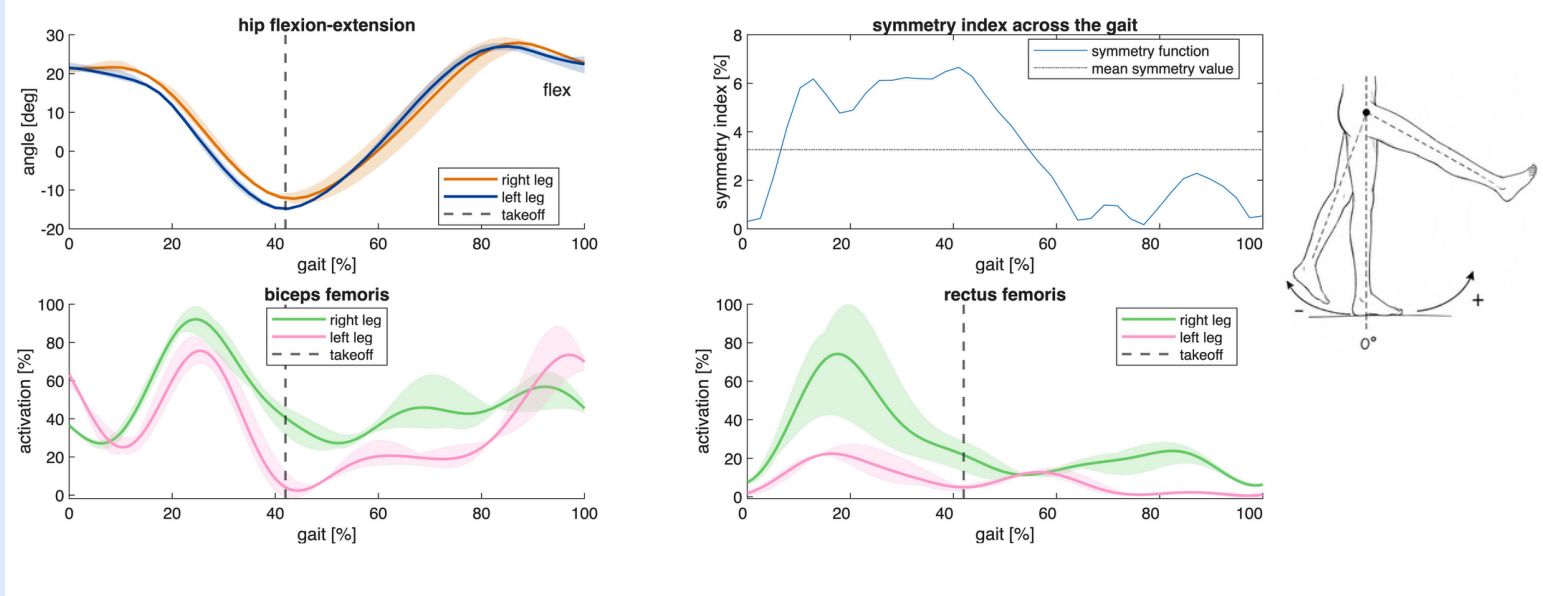
PARAMETER 1 – KNEE FLEXION/ EXTENSION



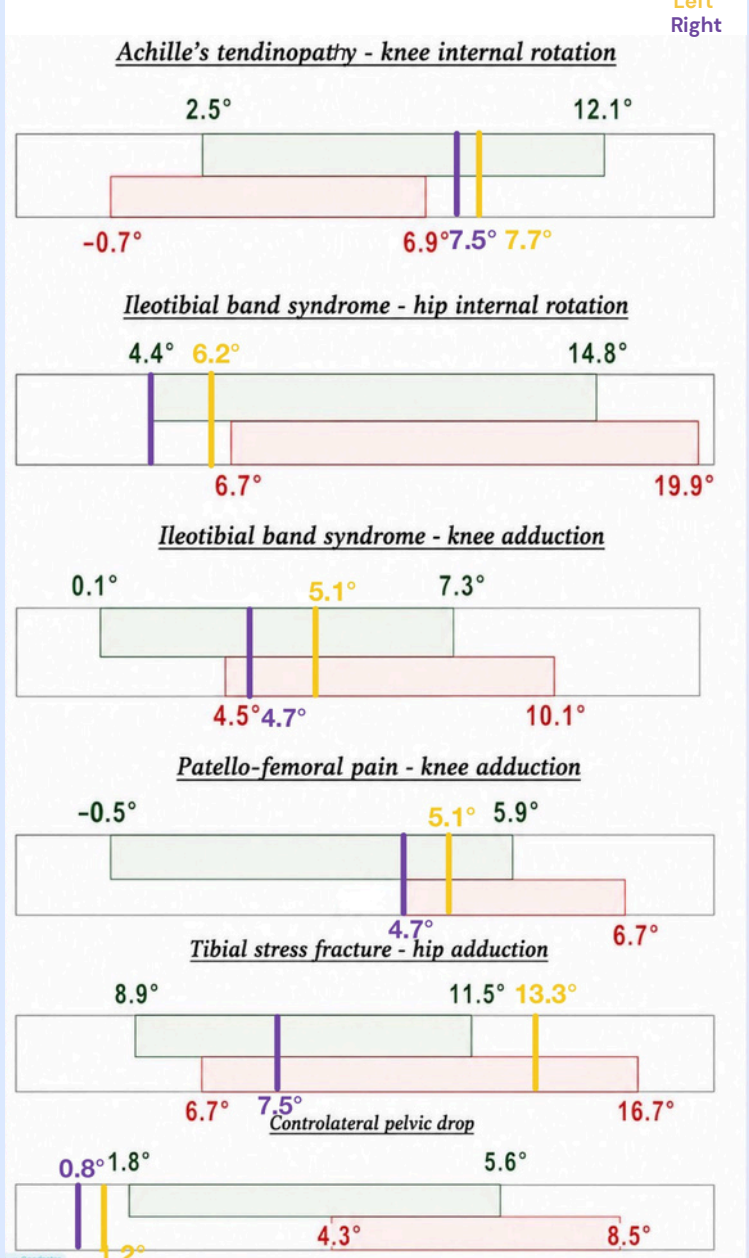
PARAMETER 2 – ANKLE DORSI/ PLANTAR FLEXION



PARAMETER 3 – HIP FLEXION/EXTENSION



INJURIES PREVENTION



PERFORMANCE

Parameter	Value
Cadence [Hz]	2.89
Contact time [s]	0.278
Step length [m]	1.255
Vertical oscillation [cm]	9.478
Trunk angle	11.27°

COMMENTS

A marked asymmetry in ankle plantarflexion/dorsiflexion is observed throughout the gait cycle . At foot strike, the left foot contacts the ground in slight plantarflexion, while the right foot shows greater dorsiflexion. This asymmetry is associated with increased activation of the right tibialis anterior. An higher activation of the gastrocnemius medialis is also observed at foot strike, although not sufficient to restore a symmetrical landing pattern. Increased activation of the right biceps femoris and rectus femoris is present as well, but no relevant asymmetries are observed at the hip or knee in the sagittal plane.

From an injury prevention perspective, a potential risk of Iliotibial Band Syndrome (ITBS) and Patellofemoral Pain (PFP) is suggested, associated with increased peak knee adduction.

From a performance perspective, a generally balanced running pattern is observed. Cadence and step length suggest a slightly greater reliance on longer steps rather than a very high step frequency. Contact time indicates a stance phase that is not particularly short, suggesting a stable but less reactive running strategy. Vertical oscillation remains moderate, with no clear evidence of excessive energy loss in the vertical direction. A slight forward trunk inclination may support forward progression and propulsion



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MILANO 1863**

BIOMECHANICS AND EVALUATION OF HUMAN PERFORMANCE

BIOMECHANICAL REVIEW OF RUNNING TECHNIQUE

NAME: Giorgio Lanzoni
AGE: 24

HIGHT 175 cm
WEIGHT 63 kg

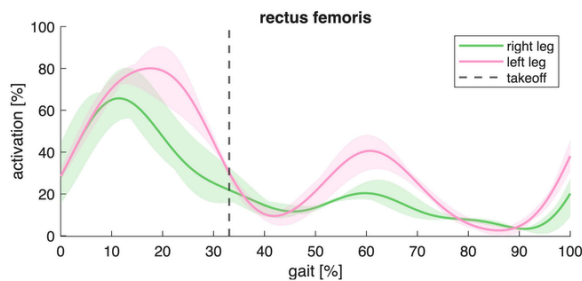
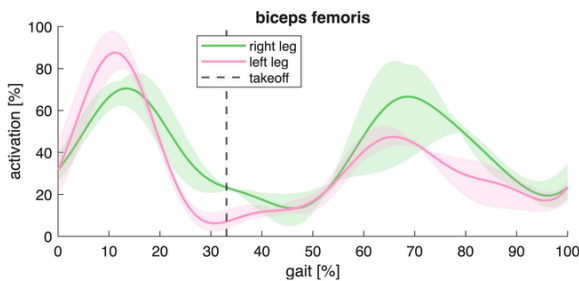
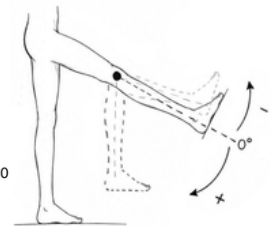
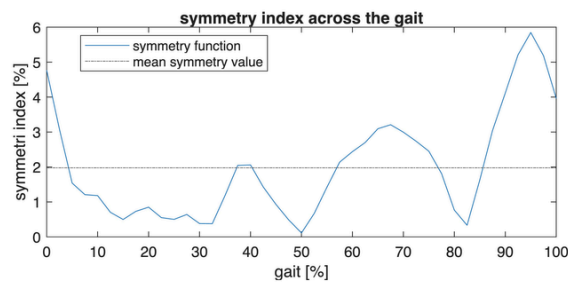
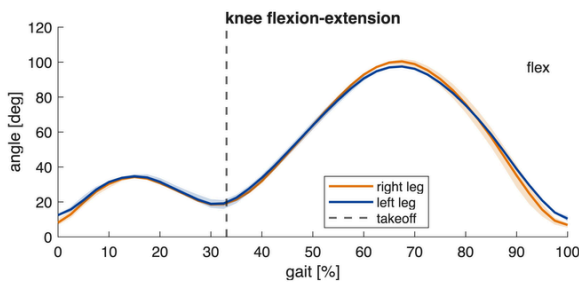
VO2MAX: 68.7 ml/Kg*min
KM/WEEK: 63

INJURIES: Microfracture of right malleolus

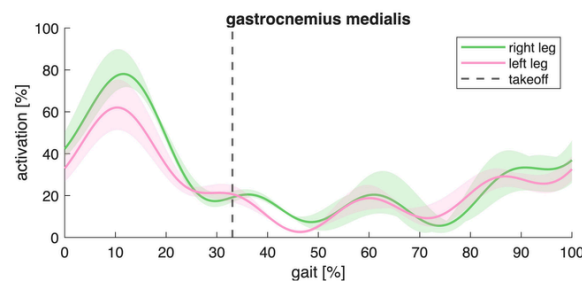
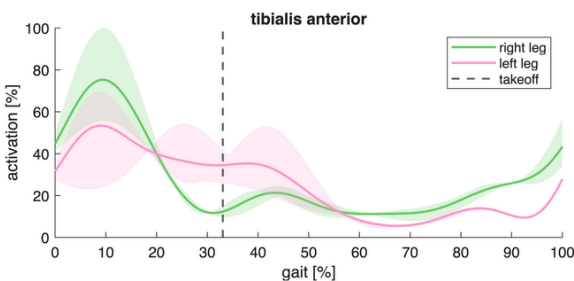
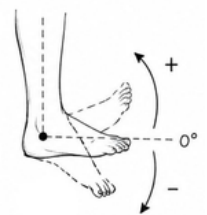
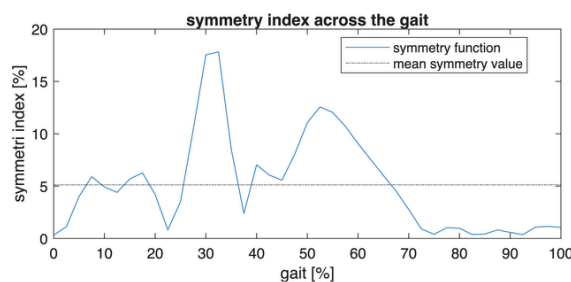
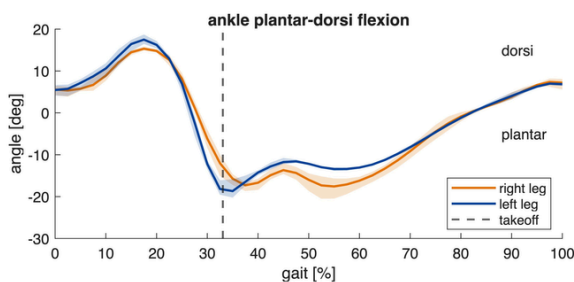
ASYMMETRIES

Evaluation of left-right asymmetry indices, focusing on kinematic parameters.

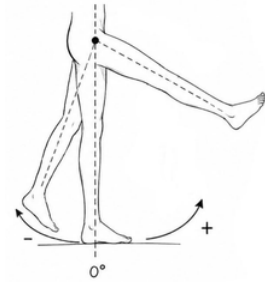
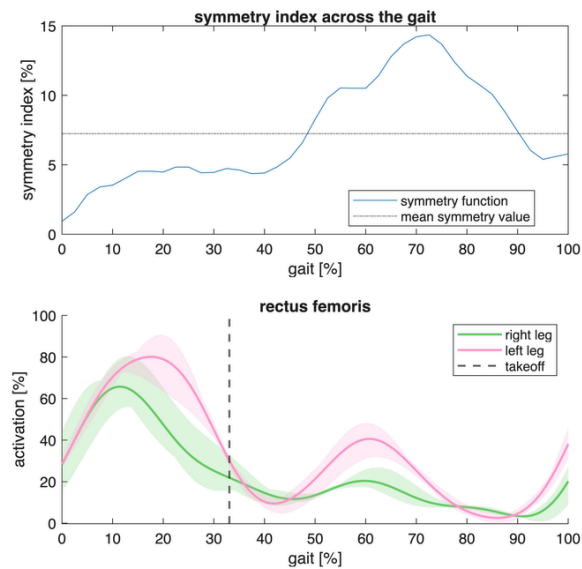
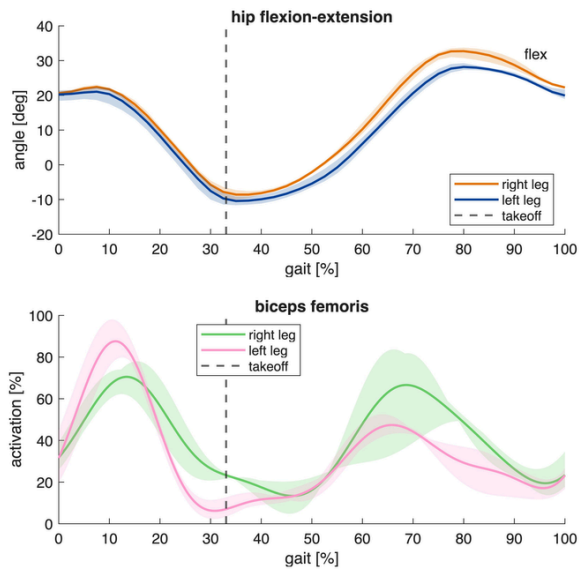
PARAMETER 1 – KNEE FLEXION/ EXTENSION



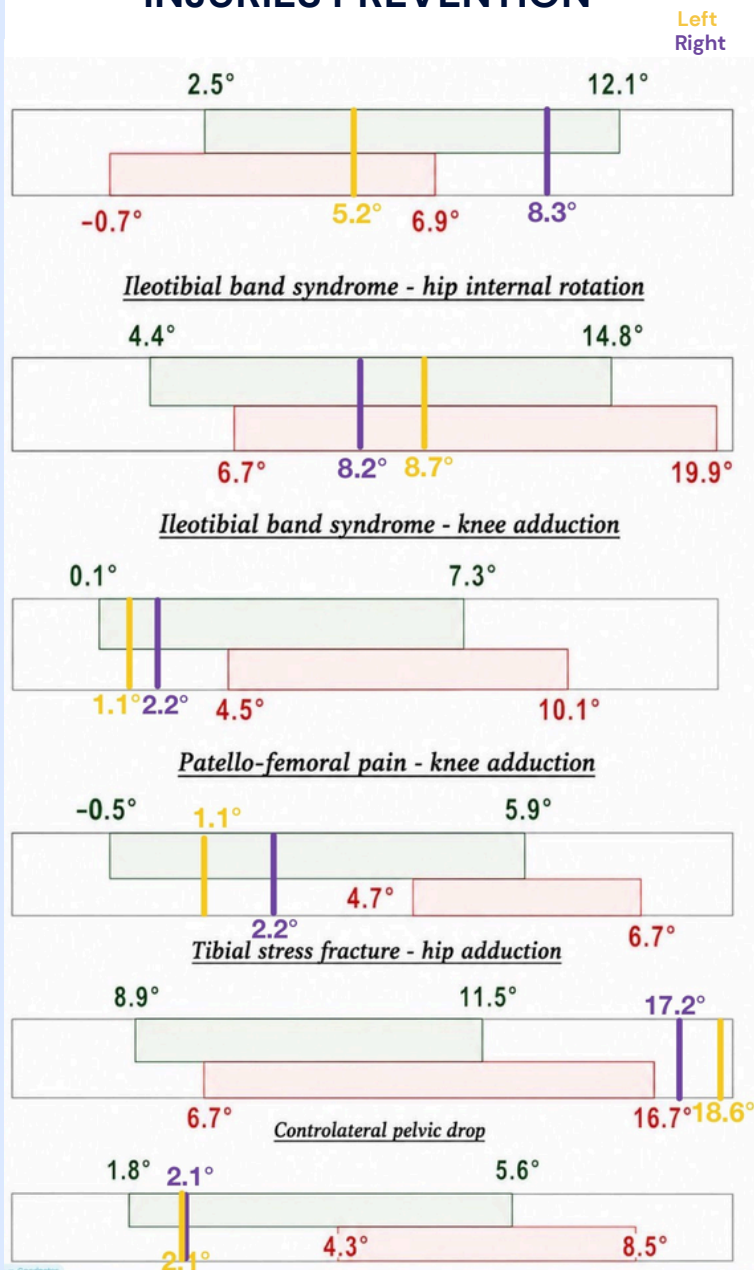
PARAMETER 2 – ANKLE DORSI/ PLANTAR FLEXION



PARAMETER 3 - HIP FLEXION/EXTENSION



INJURIES PREVENTION



PERFORMANCE

Parameter	Value
Cadence [Hz]	2.88
Contact time [s]	0.241
Step length [m]	1.249
Vertical oscillation [cm]	10.76
Trunk angle	9.65°

COMMENTS

An asymmetry in tibialis anterior activation is mainly observed during the stance phase. On the left leg, activation is lower and more prolonged than on the right leg, resulting in greater left ankle dorsiflexion during the early swing phase. Has been reported a mild discomfort in the left shin during data collection, which may be related to this pattern. Increased activation of the left biceps femoris and rectus femoris is also observed during stance, although hip and knee symmetry is generally maintained, likely due to a compensatory balance between muscle groups.

From an injury prevention perspective, a potential risk of Achilles Tendinopathy (AT) is suggested, associated with increased peak knee internal rotation compared with control values. A potential risk of Iliotibial Band Syndrome (ITBS) is also indicated bilaterally, linked to elevated peak hip internal rotation.

From a performance perspective, a generally stable running pattern is observed. Cadence and step length suggest a strategy based on moderately long steps rather than very high step frequency. Contact time is relatively short, indicating a somewhat reactive gait and a limited stance phase. Vertical oscillation is slightly increased, suggesting a partial dispersion of energy in the vertical direction. The trunk shows a moderate forward inclination, supporting propulsion and efficient running mechanics.



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BIOMECHANICS AND EVALUATION OF HUMAN PERFORMANCE

BIOMECHANICAL REVIEW OF RUNNING TECHNIQUE

NAME: Marta Crippa
AGE: 22

HEIGHT 168 cm
WEIGHT 52 kg

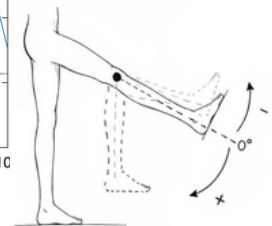
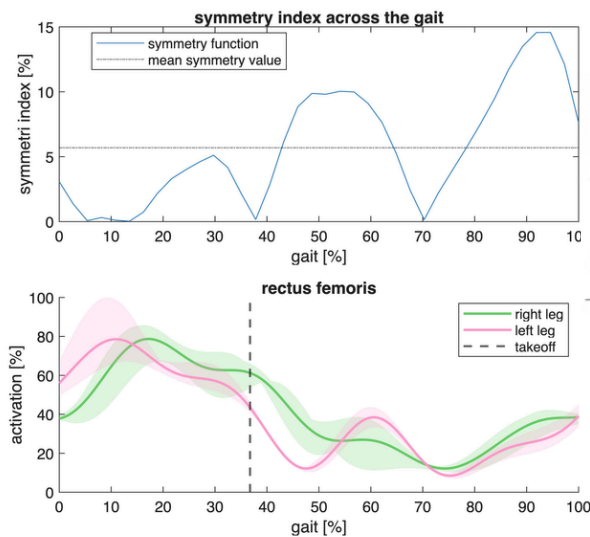
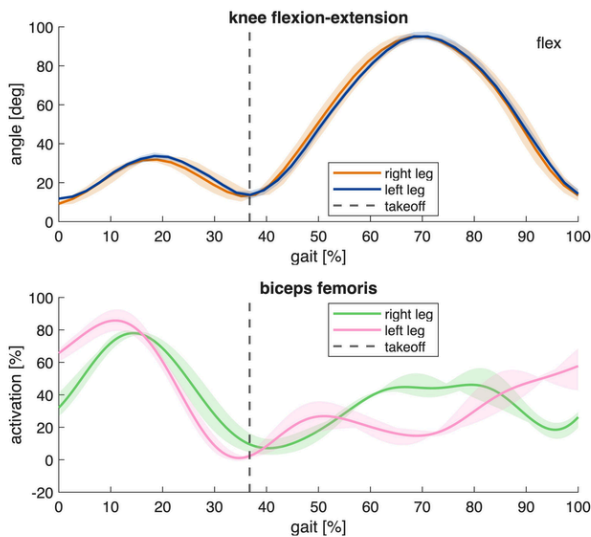
VO2MAX: 62 ml/Kg*min
KM/WEEK: 70

INJURIES: Two microfractures of the left cuboid bone

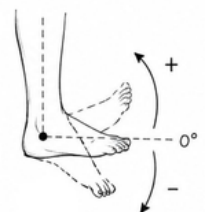
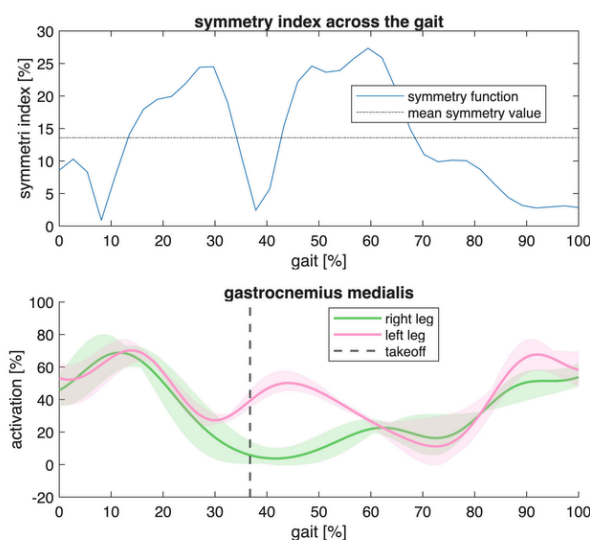
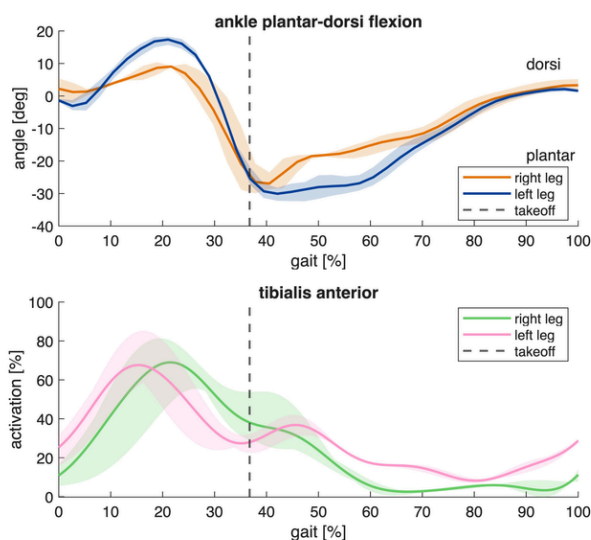
ASYMMETRIES

Evaluation of left-right asymmetry indices, focusing on kinematic parameters.

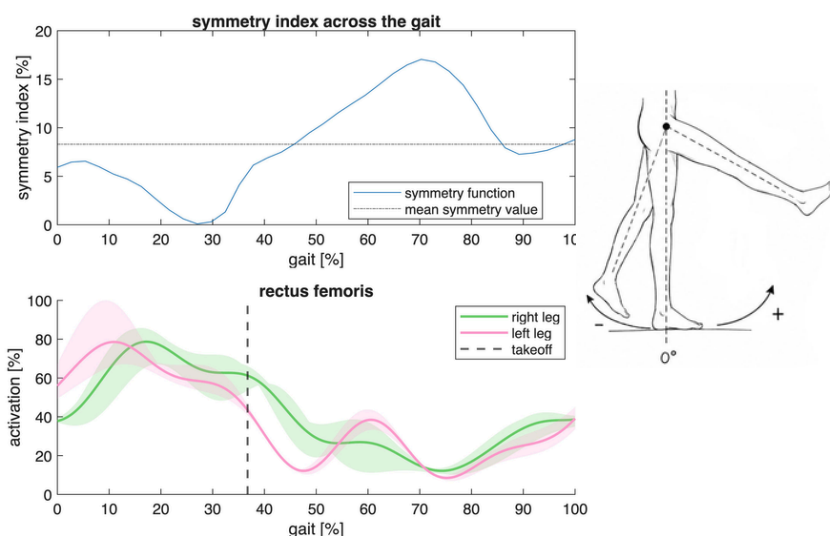
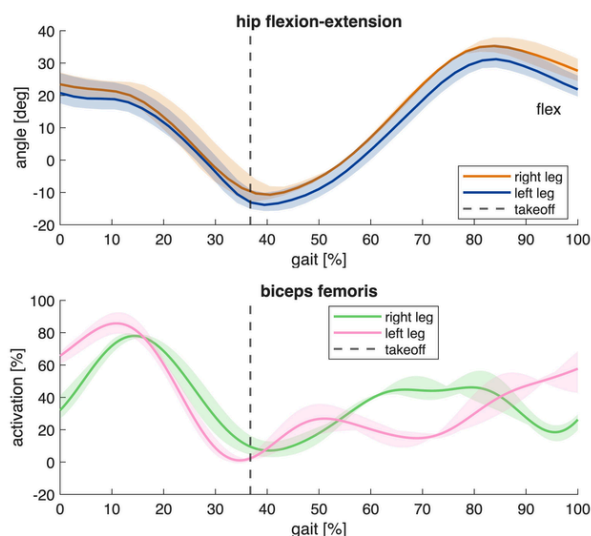
PARAMETER 1 – KNEE FLEXION/ EXTENSION



PARAMETER 2 – ANKLE DORSI/ PLANTAR FLEXION



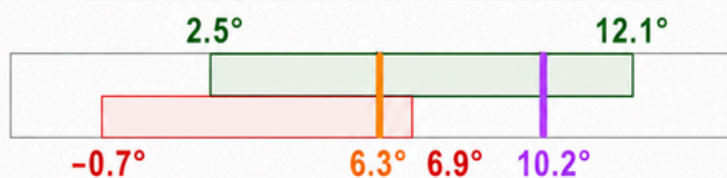
PARAMETER 3 – HIP FLEXION/EXTENSION



INJURIES PREVENTION

Left
Right

Achille's tendinopathy - knee internal rotation



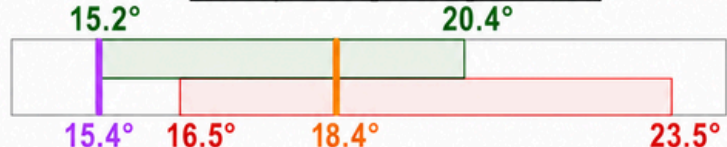
Iliotibial band syndrome - hip adduction



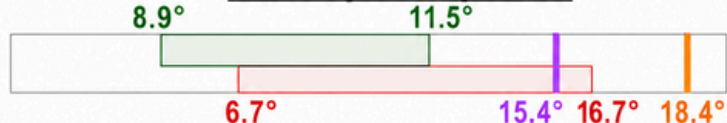
Iliotibial band syndrome - knee internal rotation



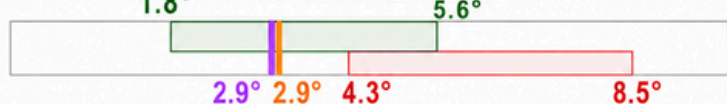
Patello-femoral pain - hip adduction



Tibial stress fracture - hip adduction



Contralateral pelvic drop



PERFORMANCE

Parameter	Value
Cadence [Hz]	3.13
Contact time [s]	0.226
Step length [m]	1.158
Vertical oscillation [cm]	9.068
Trunk angle	6.62°

COMMENTS

A symmetrical knee flexion–extension pattern is observed, associated with balanced activation of the rectus femoris and biceps femoris muscles. A slightly greater plantarflexion of the left ankle is present during early swing, likely related to increased activation of the left gastrocnemius medialis.

From an injury prevention perspective, an increased risk of Iliotibial Band Syndrome (ITBS) is suggested bilaterally due to elevated hip adduction and knee internal rotation values. The high hip adduction may also indicate a potential risk of Tibial Stress Fracture (TSF), while the left side additionally shows characteristics associated with Patellofemoral Pain (PFP). However, the limited contralateral pelvic drop may help reduce these risks.

From a performance perspective, an efficient and consistent running pattern is observed. The combination of high cadence, moderate step length, short contact time, and moderate vertical oscillation suggests a reactive running style with limited energy waste. The slight forward trunk lean (6.62°) is also compatible with efficient forward progression.



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BIOMECHANICS AND EVALUATION OF HUMAN PERFORMANCE

BIOMECHANICAL REVIEW OF RUNNING TECHNIQUE

NAME: Luigi Gianatti
AGE: 22

HEIGHT 171 cm
WEIGHT 76 kg

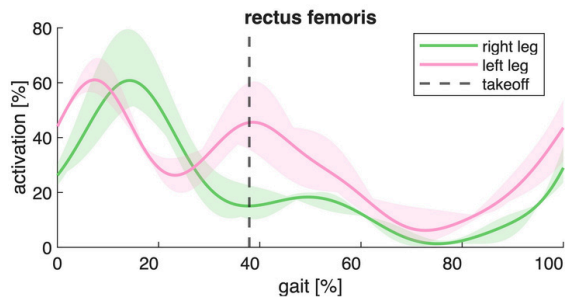
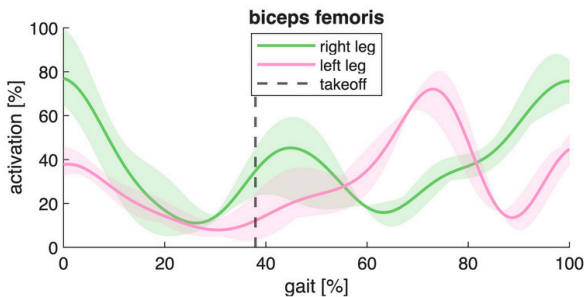
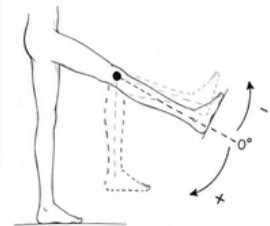
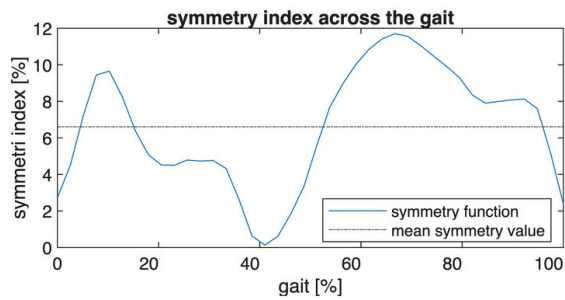
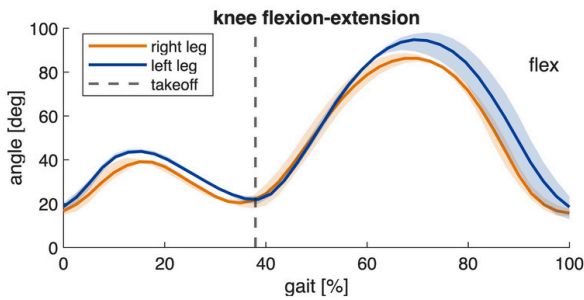
VO2MAX: 55 ml/Kg*min
KM/WEEK: 40

INJURIES: Right knee pain

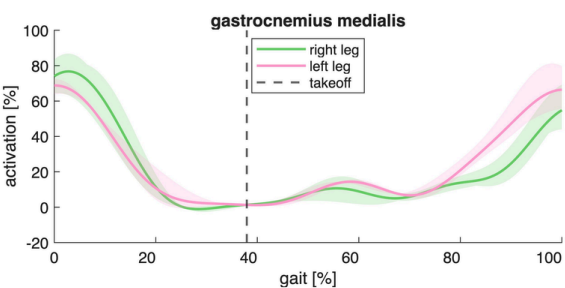
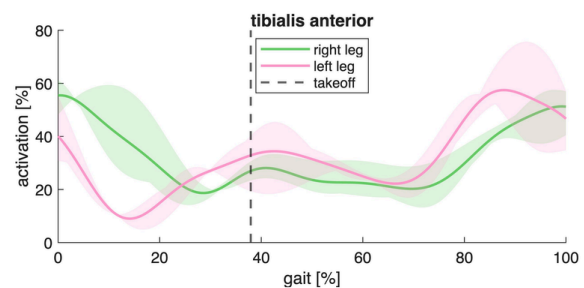
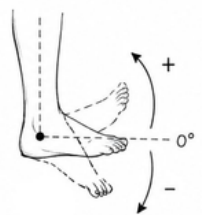
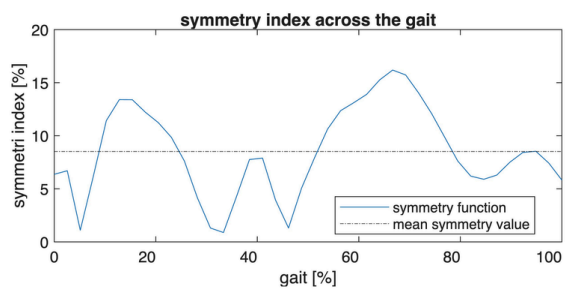
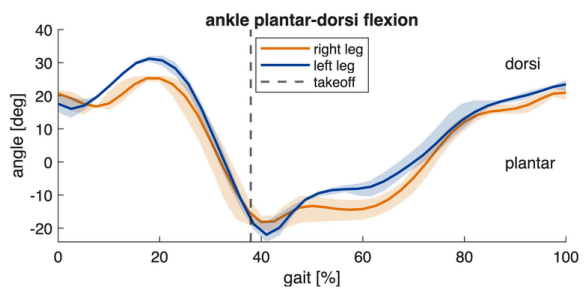
ASYMMETRIES

Evaluation of left-right asymmetry indices, focusing on kinematic parameters.

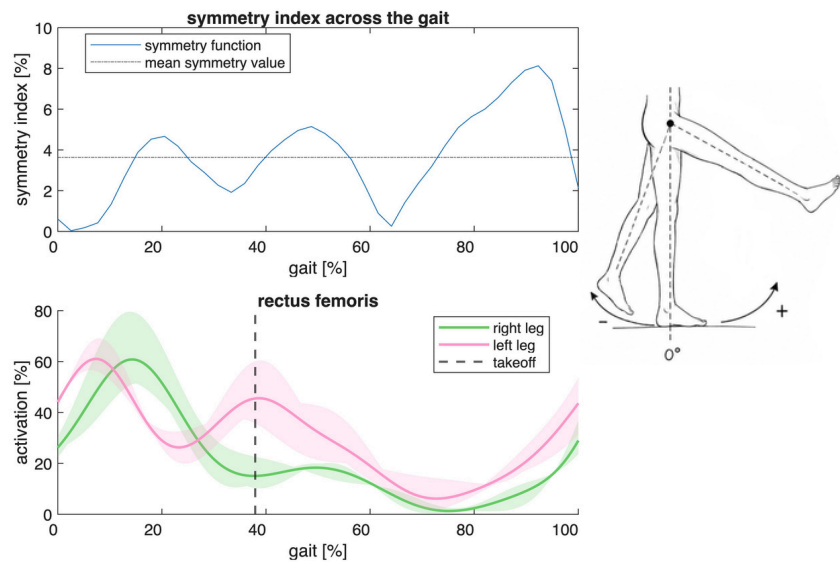
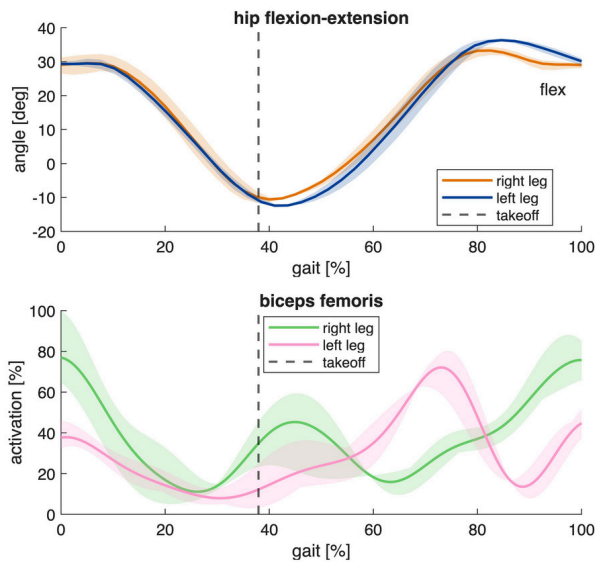
PARAMETER 1 – KNEE FLEXION/ EXTENSION



PARAMETER 2 – ANKLE DORSI/ PLANTAR FLEXION

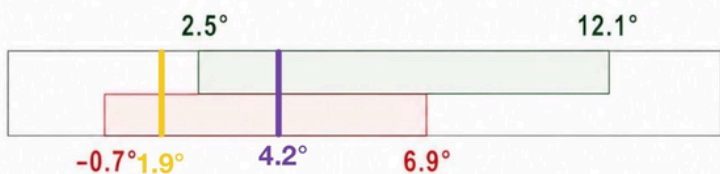


PARAMETER 3 - HIP FLEXION/EXTENSION

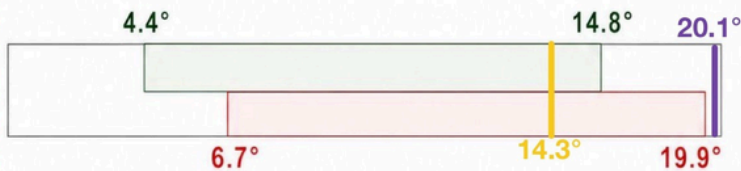


INJURIES PREVENTION

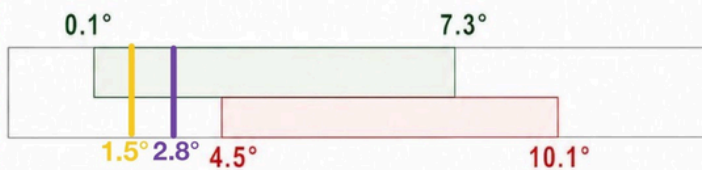
Achille's tendinopathy - knee internal rotation



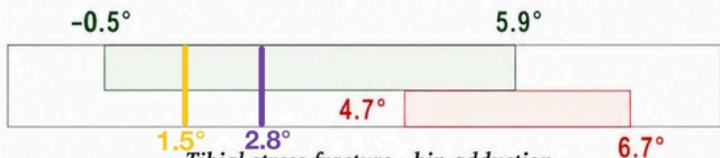
Ileotibial band syndrome - hip internal rotation



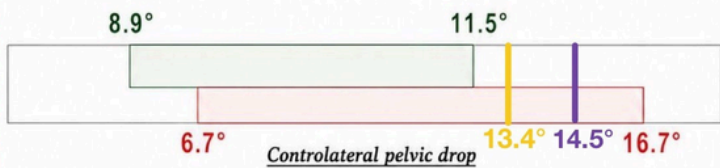
Ileotibial band syndrome - knee adduction



Patello-femoral pain - knee adduction



Tibial stress fracture - hip adduction



Controlateral pelvic drop



PERFORMANCE

Parameter	Value
Cadence [Hz]	2.90
Contact time [s]	0.267
Step length [m]	1.247
Vertical oscillation [cm]	9.79
Trunk angle	12.41°

COMMENTS

A greater left knee flexion is observed, although overall gait symmetry remains relatively preserved. Increased activation is present in the right biceps femoris during the stance phase, likely reflecting a greater stabilizing demand at the knee due to a history of ACL weakness reported during data collection. Increased activation of the right tibialis anterior is also present, possibly related to a general increase in joint stiffness associated with the injury.

From an injury prevention perspective, a high risk of Achilles Tendinopathy (AT) is indicated in the left leg and a potential risk in the right leg. A high risk of Iliotibial Band Syndrome (ITBS) is also suggested in the right leg, with a potential risk in the left leg.

From a performance perspective, a stable running pattern is observed. Cadence and step length suggest a strategy based on moderately long steps rather than very high step frequency. Contact time is slightly longer compared to previous subjects, indicating a somewhat extended but still controlled stance phase. Vertical oscillation remains moderate, with no clear evidence of excessive energy waste in the vertical direction. A more pronounced forward trunk inclination is observed, which may support propulsion and forward progression.



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BIOMECHANICS AND EVALUATION OF HUMAN PERFORMANCE

BIOMECHANICAL REVIEW OF RUNNING TECHNIQUE

NAME: Alessandro Cazzaniga
AGE: 23

HEIGHT 183 cm
WIEGHT 77 kg

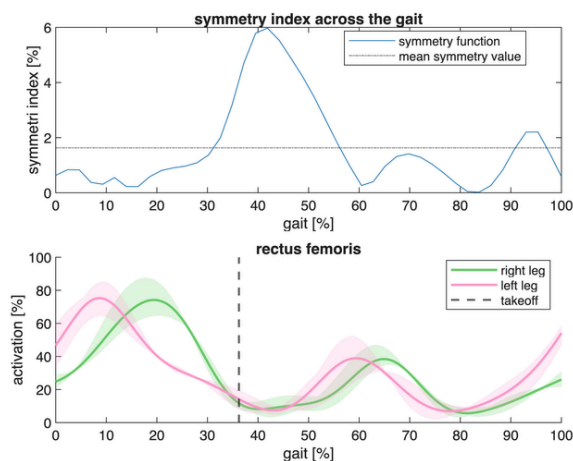
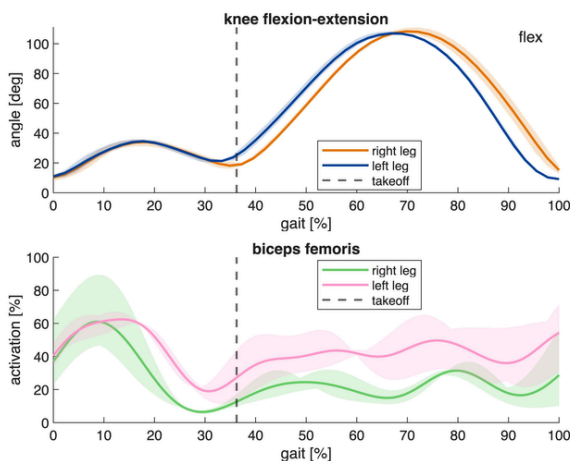
VO2MAX: 55 ml/Kg*min
KM/WEEK: 70

INJURIES: Right-sided tibial periostitis

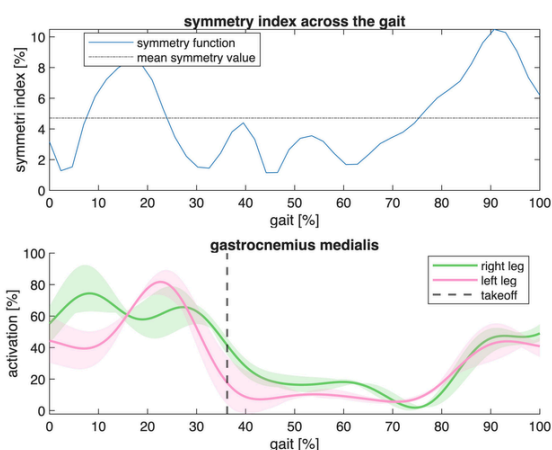
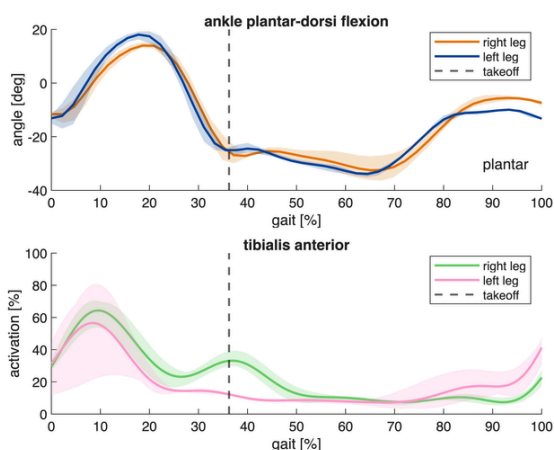
ASYMMETRIES

Evaluation of left-right asymmetry indices, focusing on kinematic parameters.

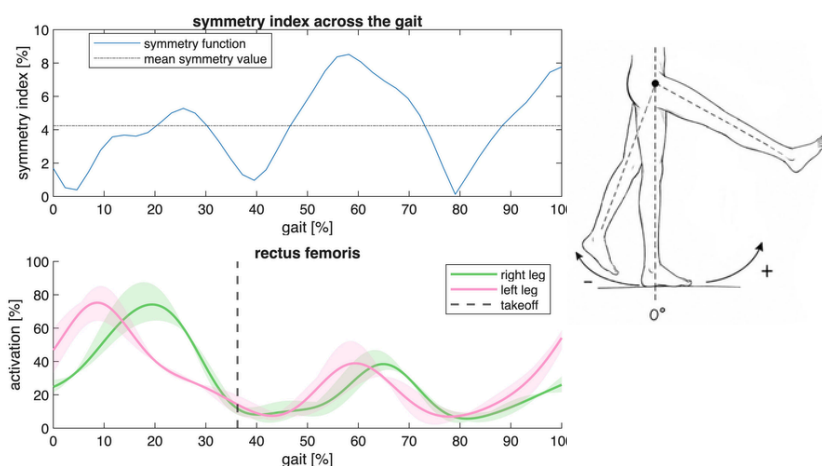
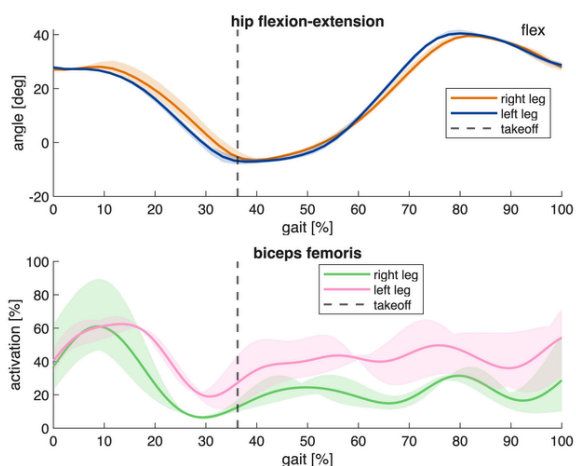
PARAMETER 1 – KNEE FLEXION/ EXTENSION



PARAMETER 2 – ANKLE DORSI/ PLANTAR FLEXION



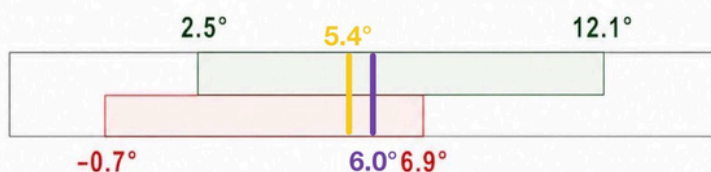
PARAMETER 3 - HIP FLEXION/EXTENSION



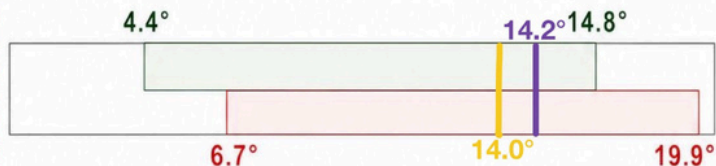
INJURIES PREVENTION

Left
Right

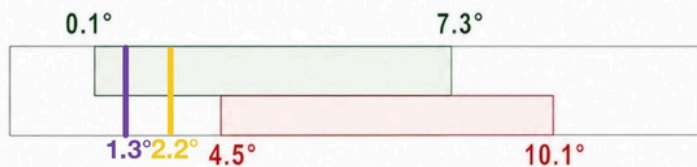
Achille's tendinopathy - knee internal rotation



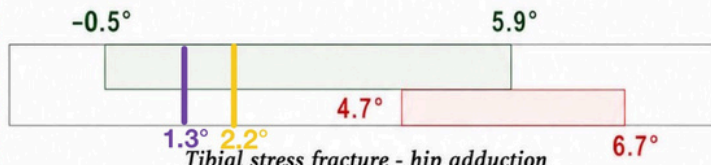
Ileotibial band syndrome - hip internal rotation



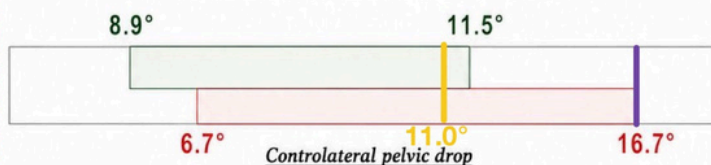
Ileotibial band syndrome - knee adduction



Patello-femoral pain - knee adduction



Tibial stress fracture - hip adduction



Controlateral pelvic drop



PERFORMANCE

Parameter	Value
Cadence [Hz]	2.63
Contact time [s]	0.264
Step length [m]	1.378
Vertical oscillation [cm]	12.58
Trunk angle	13.36°

COMMENTS

A good ankle symmetry is observed, with similar tibialis anterior and gastrocnemius medialis activation between limbs. A forefoot strike pattern is present and appears to be mainly biomechanical rather than linked to abnormal activation strategies. Slightly higher right tibialis anterior activation is observed during stance, likely related to recent recovery and strengthening after right tibial periostitis. Increased biceps femoris activation is also present across most of the gait cycle, without clear effects on hip or knee kinematics.

From an injury prevention perspective, potential risk factors are identified for Achilles Tendinopathy (AT), related to increased knee internal rotation, and for Iliotibial Band Syndrome (ITBS), related to increased hip internal rotation.

From a performance perspective, the running pattern is characterized by longer step length and lower cadence compared to other subjects. This suggests a strategy based more on stride length than step frequency. Contact time is moderate, indicating stable ground contact without a highly reactive pattern. Vertical oscillation is relatively high, suggesting some upward energy dispersion. A forward trunk lean is present and may support propulsion, although it reflects a more pronounced running posture.